Lecture 6

AI-CR-503

Digital Technology Program

Madan Bhandari University of Science and Technology (MBUST)

UNIT 6 Communication with Technical and Non-technical audience

Source Standford

Technical audience

- Technical audience are those individuals who have gained sufficient knowledge through higher education, training, or working in that field.

- These individuals are mainly involved in the development of AI algorithms, models, deployment, AI-systems etc.

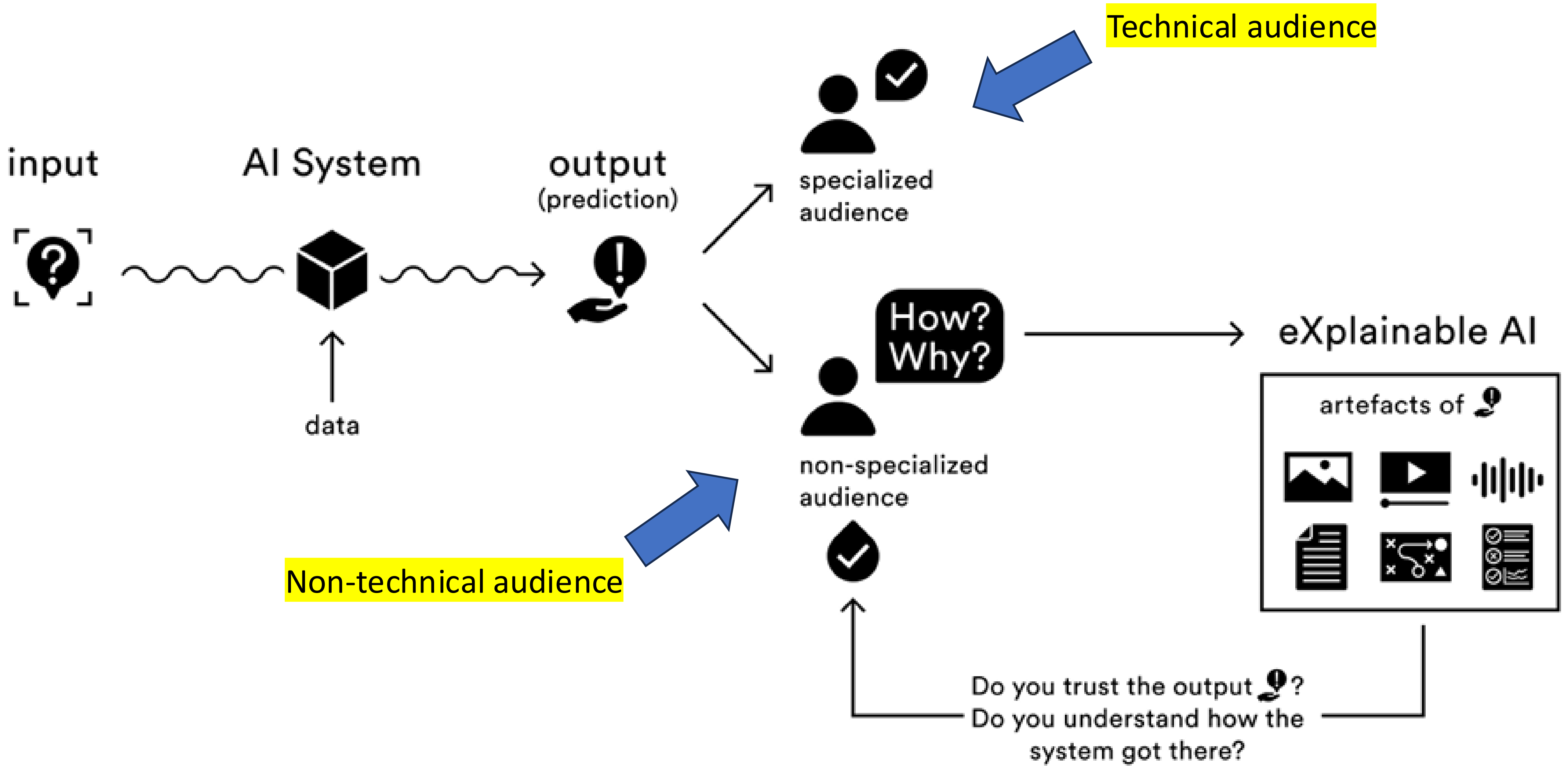
- They require specialized training or education.

Non-technical audience

- Non-technical audience are those individuals who have not studied technological subjects. These people are from other subjects than technological fields.

- These individuals are involved in non-technical roles in AI, such as advertisement, customer service, sales, business people etc.

- They lack AI literacy and require different sets of skills and training.


Technical audience
input
AI System
output
(prediction)
specialized
audience
data
How?
Why?
eXplainable AI
artefacts of
non-specialized
audience
Non-technical audience
Do you trust the output ?
Do you understand how the
system got there?

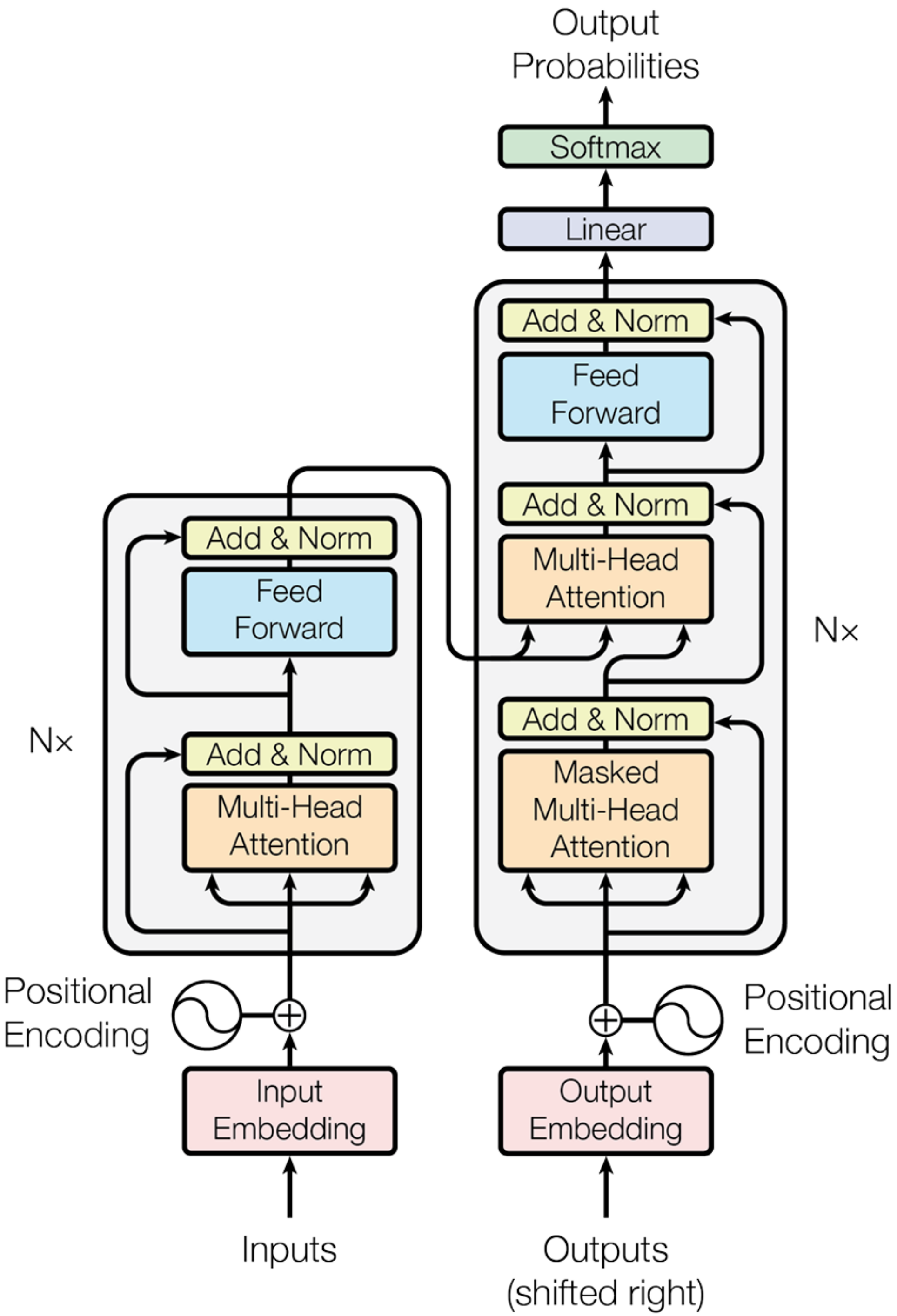
Communication with technical audience

- Introduce AI-project and its aims and objectives with correct terminology.
- State the problems and solutions.
- Explain methods such as model architecture and math behind it, datasets, data pre-processing step, training, testing, and validation steps and deployment.
- What are its limitations, dataset biases, ethics etc.
- Discuss about its end-users, trustworthiness, safety etc.
- Use examples and analogies while communicating.

Examples

- Model architecture

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$



Vaswami et al. 2017

- Language datasets.
- Useful for sequence data.
- Generative AI, Chatbot, Conversation applications etc.

Communication with non-technical audience

- Use simple language and not technical terminology.
- Tell them about its usefulness and risks.
- Relate the AI with analogies and demos.
- General discussion with the audience.

Examples

1. Explainable AI (XAI)

2. AI literacy

1. Explainable AI (XAI)

- A myriad of XAI methods have been proposed over the years to provide users with explanations of how the models made the decisions.
- These include SHapley Additive exPlanations (SHAP), which aims to quantify feature importance, shedding light on how each feature contributes individually and in conjunction with others to produce the results given by the model.
- Another model-agnostic method proposed is the Local Interpretable Model Agnostic Explanations (LIME), which uses local, smaller models to explain individual predictions and highlight feature influence.

- Explanations using images have also been tackled, with approaches such as Grad-CAM, applying highlights in images to show the importance of specific details for the prediction.
- Regarding Natural Language Processing, previous research has attempted to infer the importance of words and how they influence the sequence generated. Some of the most commonly applied visualizations for Transformers– one of the most popular model architectures– include attention maps and other plots that contain information on a given characteristic of the model.

2. AI literacy

- Long and Magerko's AI Literacy Framework.
    a) What is AI?

    b) What can AI do?

    c) How does AI work?

    d) How should AI be used?

    e) How do people perceive AI?

Thank you.